

NOMAD PIZZA | System Requirements Document

(Public, administrative, and electronic ticket interfaces)

Nomad Pizza provides a public ordering interface, a protected administrative interface, and a public ticket view. The menu, cart, and order summary share components and Firestore data; administrative navigation depends on the user session and profile permissions.

- Public interface (index.html): lets customers browse the menu, select products, review the cart, submit orders, and open the ticket through its link or QR code.
- Administrative interface (admin.html): requires authentication. It reuses the ordering flow and adds navigation to active orders, menu settings, sales, inventory, password changes, and user management according to the user profile.

1. General Architecture

1.1 Backend and persistence

- Persistence: Cloud Firestore in the nomadpizza database. The data model uses products, ingredients, settings, counters, orders, users, inventory, and purchases.
- Local environments: the app served by the Firebase Hosting Emulator at localhost:5000 connects to the local Auth and Firestore emulators (ports 9099 and 8080). Vite on its development port does not enable that routing and keeps the configured Cloud Firebase connection.
- Authentication: Firebase Authentication with email and password for admin.html. The public interface and ticket do not require sign-in.
- Real-time updates: users with permission to dispatch orders receive the pending-order count. The ticket listens to the order document to reflect status changes.
- **Dynamic catalog:** unlike a fixed catalog, products, categories, ingredients, and extra-ingredient prices are managed in Firestore and loaded dynamically in both interfaces.

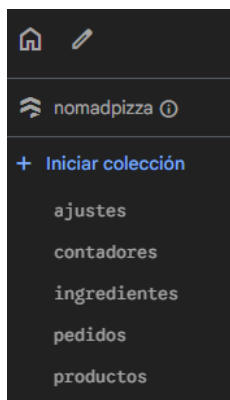


Figure 1. Cloud Firestore database.

Identificador	Proveedores	Fecha de creación ↓	Fecha de acceso	UID del usuario
fersingh.fs@gmail.com	✉	1 sept 2026	13 sept 2026	KID6NF9QtISYq21cKd3AuQx7...
jan_horacio@hotmail.c...	✉	1 sept 2026	14 sept 2026	fo9ZWEIdHLYTlcTlqVrja6UBrf...

Figure 2. Authenticated users in Firebase Authentication.

2. Public Interface (Customer Ordering)

Available in index.html. It includes the Menu, Cart, and Order Information screens, with a header and cart counter. The ticket is a separate page in pedido.html, served at /pedido/{orderId}. The public interface does not display the administrative sidebar.

2.1 Menu Screen

Order type selector

- The “Order type” selector offers “Dine in” and “Takeout”; the selected option is stored with added products and shown in the cart and ticket.
- When the order type changes, prices for products affected by this setting (for example, Pizza Dog) are recalculated live and the corresponding promotion badge is shown or hidden.

Categories and product order

Products are grouped by the type field and displayed in this order: Specials, Pizzas, Other, Drinks, and Desserts. Singular type names are normalized, and unrecognized categories appear under Other.

- Within each category, temporary/special products are listed first.
- A product is visible only if it is not marked as hidden and, when it has scheduled dates, the current date and time fall within its configured start and end dates.

Standard product card

Applies to any non-customizable product (drinks, desserts, fixed pizzas, specials without ingredients, and similar items). Each card includes:

- Optional image (blurred background with a sharp image layered on top); if the product is a special, a badge appears with its configured text (for example, “Pizza of the Month,” “Special of the Day,” or a generic badge type).
- Product name and description.
- Price (fixed, or calculated dynamically for Pizza Dog; see the special rules below).
- Quantity counter to increase/decrease the quantity (range 0–99).
- “Add” button that adds the selected quantity to the cart and resets the counter to 0.

Customizable product card

Products marked as customizable in Menu Settings display ingredient selectors and follow the maximum configured for each product.

- Optional image and product description.
- **Dynamic price:** the product base price covers the first ingredient, plus the extra-ingredient price (configurable in Menu Settings) for each additional ingredient.
- A default ingredient dropdown, with an “Add Ingredient” button that adds more selectors up to the product’s configured maximum (the button is automatically hidden at the limit).
- Each selector prevents an ingredient already chosen in another selector for the same product from being selected again (available options stay synchronized).
- A single quantity counter (0–99) applies to the selected ingredient combination.
- “Add” button that adds the selected quantity and ingredient combination to the cart, then resets the selector block to one ingredient and the counter to 0 without changing products already in the cart.
- Available ingredients come from the ingredients collection in Firestore (see Menu Settings); an ingredient hidden by an administrator no longer appears in the selectors.

Special rules — Pizza Dog

- When “Dine in” is selected: the price is fixed at \$60 per unit, with no promotion.
- When “Takeout” is selected: a 2 for \$100 promotion applies (\$50 per unit in pairs); any remaining unit that does not complete a pair is charged at \$60.

- The promotion is applied automatically based on the quantity entered; the customer does not need to select it manually.
- A “2 for \$100 PROMO!” badge appears next to the product name only when “Takeout” is selected.

Temporary specials and badges

- A product can be configured as a temporary special with an end date (expiration) and, optionally, a start date (scheduled future launch).
- It can be assigned custom badge text (for example, “2 for \$100!”) and a predefined badge type: Special, Promotion, or Limited Edition.
- It is displayed only while the current date and time fall within its active period.
- It behaves like a standard product card (optional image, description, price, counter, and Add button).

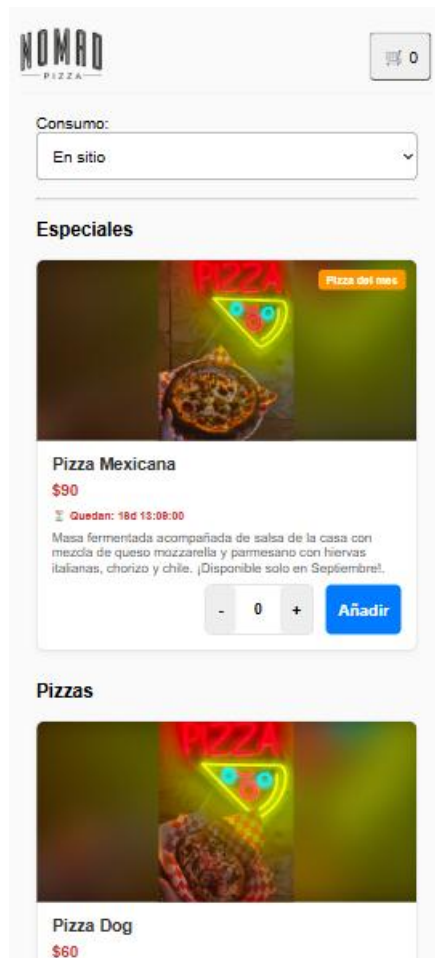


Figure 3. Pizza Nomad menu (public interface).

2.2 Cart Screen

Customer name field

- The Name field is required to submit an order and accepts up to 18 characters.
- It accepts letters and spaces only (numbers and special characters are filtered using a regular expression).
- The first letter is automatically capitalized.
- Maximum length: 18 characters.

Changing quantities in the cart

- Each grouped product has “-” and “+” controls to adjust the quantity; the line subtotal and cart total are recalculated.

- When the customer tries to decrease a quantity from 1 to 0, an in-page dialog asks whether to remove the product.
- If confirmed, the line is removed; if the cart becomes empty, the Menu screen is displayed again.
- If canceled, the line keeps its original quantity.
- Each line also has a separate “Remove” button that removes the product without asking for confirmation.
- If the cart becomes empty after either action, the system automatically returns to the Menu screen.

Information shown for each line

- Product name and base description (from the catalog).
- Additional details when applicable: selected ingredients for customizable products; no detail is displayed when the value is “Regular” or “Promotion.”
- An order type badge (“Dine in” / “Takeout”) corresponding to that specific line.
- Line subtotal and the order total at the end of the list.

Cart grouping logic (groupCart)

Before display, the internal cart is reorganized into grouped lines according to these rules:

- **Pizza Dog:** items are grouped by order type. Dine-in items always appear in one “Regular” line at \$60 each. For takeout, pairs appear in a “Promotion” line (\$100 per pair, or \$50 each), and any leftover odd unit appears separately as “Regular” at \$60.
- **Customizable products:** items are grouped by product + exact ingredient combination + order type. Different ingredient combinations (or different ingredient counts) produce separate lines, even for the same product. Unit price is calculated as base price + (number of ingredients – 1) × extra-ingredient price.
- **Standard products (drinks, desserts, and other pizzas):** items are grouped by product + order type. Drinks display “Drink” as the description; other items use “Regular” unless they have their own detail (for example, a special label).

Implementation note: the cart summary calculates the extra charge using the Menu Settings value; some menu price views still use a fixed \$16 formula per additional ingredient. If the setting changes, verify that the price shown before adding an item matches the cart price.

Payment method

The list allows customers to select Cash, Card, or Bank Transfer.

- Cash: no additional section is shown in the cart.
- Bank Transfer: the cart and order summary show the bank, beneficiary, CLABE, and instructions for sending the receipt through WhatsApp.
- Card: no additional payment information is shown in the cart.

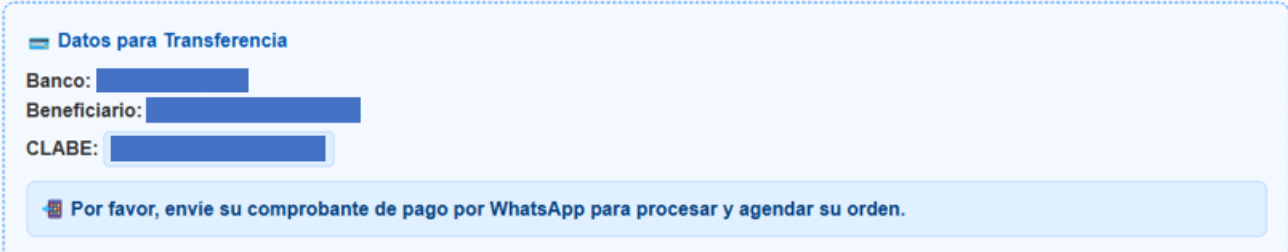
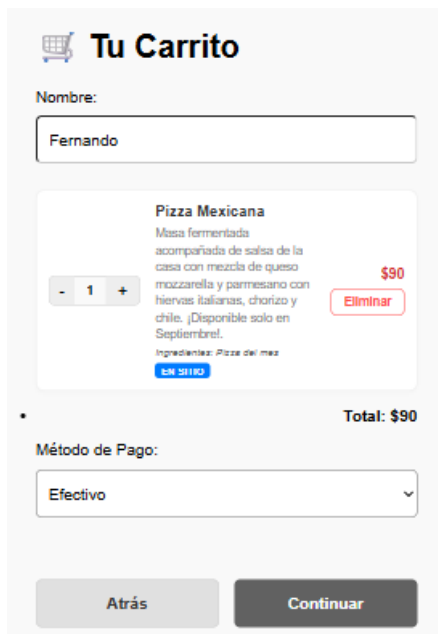


Figure 4. Bank transfer information.

Buttons

- “Continue”: validates that the name is not blank, saves the order, and opens Order Information. In admin.html, an additional in-page dialog lets the user choose between viewing the summary and going to Active Orders.

- “Back”: returns to the menu without submitting the order. The header button lets the customer return to the cart when it contains items.
- The cart button/icon in the header opens the Cart screen.



Tu Carrito

Nombre:
Fernando

Pizza Mexicana
Masa fermentada acompañada de salsa de la casa con mezcla de queso mozzarella y parmesano con hiervas italianas, chorizo y chile. ¡Disponible solo en Septiembre!
Ingredientes: Pizza del mes
\$90
Eliminar

1

Total: \$90

Método de Pago:
Efectivo

Atrás Continuar

Figure 5. Cart screen.

2.3 Order Information Screen

Order ID generation

- Format: NO-YY-DDD-NNN (example: NO-26-250-001), where YY is the two-digit year, DDD is the three-digit day of the year, and NNN is the order sequence for that day.
- The daily sequence is retrieved and incremented inside a Firestore transaction on counters/{prefix}, preventing duplicate identifiers when orders are submitted concurrently.
- All products in one order share the same order ID.
- The order is saved in Firestore with an initial status of “Pending.” The transaction also updates the counter and deducts inventory components when corresponding recipes are configured.

Information displayed

- Order ID.
- Customer name.
- Order product details (quantity, name, ingredients/details when applicable, order type, and subtotal).
- Selected payment method.
- Order total.

Button

“Return to Menu”: clears the cart and customer name, then returns to the Menu screen in its default state.

QR code

“Scan or click to view your ticket.” Clicking or scanning the QR code takes the user to the Electronic Ticket screen.

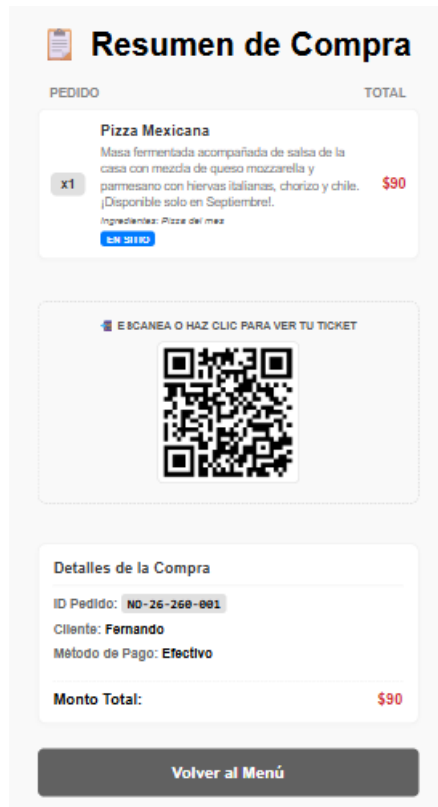


Figure 6. "Order Summary" screen.

2.4 Electronic Ticket Screen

The ticket is a separate public page, not an internal panel navigation screen. It displays the current order status in real time and the payment details for the selected method.

Information displayed

- Order ID.
- Date and time.
- Order product details (quantity, name, ingredients/details when applicable, and subtotal).
- Order total.
- Customer name.
- Selected payment method.
- Order type.
- For cash payments, the ticket includes fields for the amount received and change; if these values are not saved with the order, it displays \$0.
- For bank transfers, it displays bank details and a WhatsApp link for sending the receipt. On the ticket, these details are defined in `pedido.html/src/pedido.js`.
- For card payments, no additional payment information is displayed.
- Order status in real time.
- "Thank you for choosing us! 😊"



Figure 7. “Electronic Ticket” screen.

3. Administrative Interface

Available in admin.html. After sign-in, it retains the shared Menu, Cart, and Order Information flow and adds administrative screens. Sidebar options depend on the user’s role and permissions; Firestore rules enforce data access.

3.1 Authentication and security

- **Sign-in:** email and password through Firebase Authentication. Access is controlled by user account.
- Until a user signs in, only the login screen is displayed; protected content remains hidden.
- Automatic inactivity sign-out: after five minutes without mouse, keyboard, click, or scroll activity, the session is closed and a notice is displayed.
- The authenticated user’s email and role are shown at the top of the sidebar.
- Manual sign-out (the “Sign Out” button) clears the local cart and closes the sidebar.

Change Password screen

- A form with one field: new password (required, at least 6 characters).
- When submitted, it updates the authenticated user’s password in Firebase Authentication.
- If Firebase requires a recent sign-in for security, a notice asks the user to sign out and sign in again before changing the password.



Inicio de Sesión Admin

Ingresar tus credenciales.

Correo:

Contraseña:

Ingresar

Figure 8. “Admin Sign-In” screen.



Cambiar Contraseña

Ingresar tu nueva contraseña para actualizarla en Firebase Auth.

Nueva Contraseña (mínimo 8 caracteres)

Actualizar Contraseña

Figure 9. “Change Password” screen.

3.2 Sidebar

- It opens using the menu icon (☰) in the header and is available on every protected panel screen.
- It closes using the “X” button or automatically when a navigation option is selected.
- Visible options depend on the user’s profile and permissions:
 - 🍷 Take Order — opens the menu when the profile has the take_orders permission.
 - 📄 Active Orders — opens pending orders when the profile has the dispatch_orders permission.
 - 📊 View Sales — opens the sales history when the profile has the view_sales permission.
 - ⚙️ Menu Settings — opens product and ingredient management when the profile has the edit_menu permission.
 - 🔑 Change Password — available to authenticated users.
 - 👤 Manage Users — administrators only; allows them to register collaborators, assign and update roles, and deactivate accounts.
 - 🚪 Sign Out — ends the authenticated session.

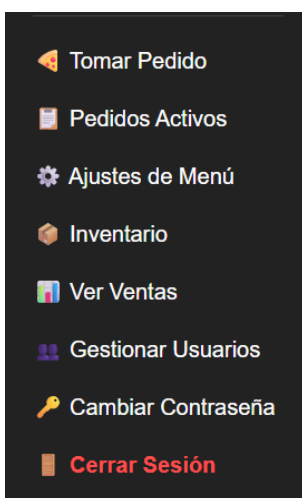


Figure 10. Sidebar navigation options.

3.3 Administrative Header

- Hamburger menu button (≡) to open the sidebar.
- Clickable centered logo that returns to the menu.
- “📄 **Orders**” button with a counter: displays the real-time number of orders with “Pending” status. When at least one order is pending, the button turns red and its text becomes bold.
- Cart button (🛒) with item count, as in the public interface.



Figure 11. Pizza Nomad header (admin interface).

3.4 Active Orders Screen

This screen is exclusive to the administrative panel; it is not available in the public interface. It manages incoming orders.

- Only orders with “Pending” status are displayed.
- A button toggles the list order between “Newest first” and “Oldest first,” based on the actual order date and time.
- Each order card displays the order number, item details (quantity, name, ingredients when applicable, order type), customer name, payment method, and total.
- Actions for each order: ✓ Delivered changes the status to “Order delivered,” and ✕ Cancel changes it to “Order canceled.” After the status is updated, an in-page message appears and the order leaves the pending list.
- New-order notification: an in-page dialog offers “View order information” and “Go to Active Orders.” The first opens the summary; the second opens the pending-order list.



Figure 12. “Active Orders” screen.

3.5 Menu Settings Screen

The screen contains the extra-ingredient price setting and two management areas: Products and Ingredients. It also includes inventory recipe settings in the product form.

Tab 1 — Products

The “New Product” form contains these fields:

- **Category (required):** Pizzas, Drinks, Desserts, Other, or Specials (Temporary).
- Drink subtype (shown only when the category is Drinks): Fresh Water, Soft Drink, or Other.
- Product name (required).
- Description (required).

- Price in pesos (required).
- “Has image?” toggle: when enabled, it displays a field for the image URL.
- “Customizable?” toggle: when enabled, it displays a numeric field for the maximum number of ingredients (defaults to 3 if left blank).
- “Schedule future launch? (Start)” toggle: when enabled, it displays the product availability start date and time; only then is the following end/expiration toggle enabled.
- “Schedule End (Expiration)” toggle: when enabled (or when the selected category is “Specials”), it displays the “Specials Settings” section with custom badge text (for example, “2 for \$100!”), badge type (Special / Promotion / Limited Edition), and expiration date and time.
- **Automatic identifier generation:** a product ID is generated from its name (lowercase, without accents or special characters, removing linking words such as “the,” “of,” “with,” “a,” and “and,” then joining the remaining words with underscores).
- “Save Product” button: saves the product to Firestore. When editing a product, if its new name generates a different ID, the previous document is deleted and a new one is created. After saving, the form is cleared, the list and menu are updated, and the Menu screen is displayed.

Product Management (list)

- Lists all products with their name and category.
- “Edit” button: loads the product data back into the form for editing.
- “Hide” / “Show” button: toggles product visibility in the menu without deleting it (sets hidden = true/false).
- “Delete” button (X): asks for confirmation and permanently deletes the product from Firestore.

Figure 13. “Menu Settings” screen — Tab 1, “Products.”

Tab 2 — Ingredients

- Text field and “Add” button to create a new ingredient in Firestore.
- Ingredient list with a “Hide” / “Show” button (hides the ingredient from menu selectors without deleting it) and a delete button (X) with confirmation.
- **Extra-Ingredient Price:** a numeric field (pre-filled with the current value) and a “Save Price” button that updates the price charged for each additional ingredient on customizable products in Firestore. Empty values and values less than or equal to zero are rejected. After saving, the menu reloads to reflect the new price.

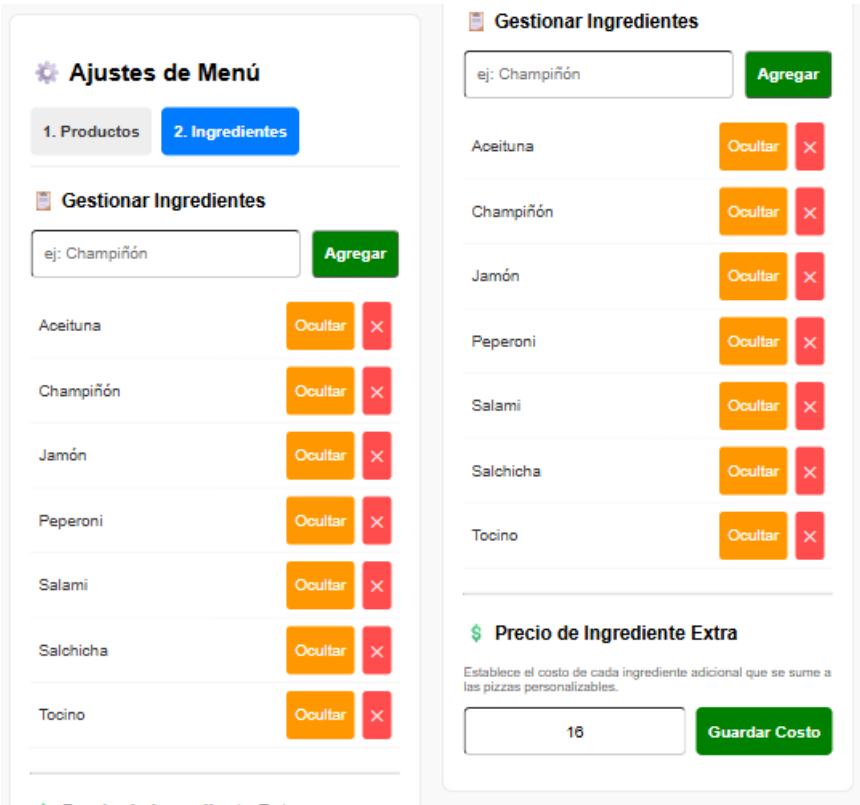


Figure 14. “Menu Settings” screen — Tab 2, “Ingredients.”

3.6 View Sales Screen (History / Database)

Displays sales loaded for the selected period, broken down by product. Each row retains the order, payment, and status context.

Table columns

Table 1. Columns in the “View Sales” screen.

Column	Description
Date	Date the order was submitted, based on the device used. Format: YYYY/MM/DD.
Time	Time the order was submitted, based on the device used. 24-hour format.
ID	Order identifier (may repeat when an order contains multiple products).
Order Type	Dine in or Takeout.
Customer	Customer name saved with the order.
Qty	Quantity sold for that product/combination in the order.
Product	Grouped product name (for example, including the ingredient count for customizable products).
Description	Product detail: special theme name, selected ingredients for customizable products, or “Regular/Promotion” for Pizza Dog.
Revenue	Amount sold for that grouped line.
Payment	Payment method used for the order (Cash, Card, or Bank Transfer).
Status	Pending, Order delivered, or Order canceled.

Table functions

- **Excel-style filters:** each column has a text field below its heading that filters live by partial match (case-insensitive); all active filters are combined.

- **Sort by column:** clicking a heading sorts ascending/descending. Date and time are sorted chronologically, quantity and revenue numerically, and other fields alphabetically.
- **Month and year filter:** displays only order records from the selected period. Records can be filtered for any year with saved data.
- “Reset Filters” button: clears all text filters and the current sort order.
- **“Export CSV” button:** downloads an Excel-compatible .csv file named Report_NomadPizza_DD-MM-YYYY.csv.
- The text color in the Status column indicates the order state: green for delivered, red for canceled, and orange for pending.



Figure 15. “Sales History” screen.

Observed behavior: text filters and sorting apply to the records loaded in the table. CSV export is generated from the loaded sales dataset, not from the filtered results shown on screen.

3.7 Manage Users Screen

Available only to the administrator profile. It includes the “Register New” and “Manage Permissions” tabs.

Register a collaborator

Requests a name, email, and initial role: Waiter, Kitchen, Manager, or Administrator. When registered, it creates an Authentication account and saves the profile with initial permissions in users.

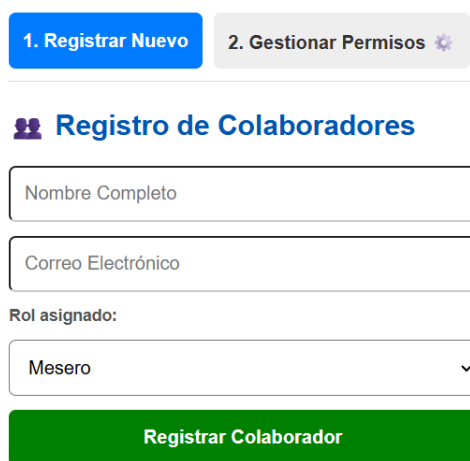


Figure 16. “Register Collaborators” screen.

Manage profiles

Lists collaborators with their name, email, and role. The administrator can change another user’s role and save; the change also updates that user’s permissions. The administrator can deactivate other collaborators. The currently signed-in administrator account cannot be modified or deleted from this list.

Current permissions by role

Waiter: take and dispatch orders. Kitchen: dispatch orders. Manager: take orders, dispatch orders, manage inventory, view sales, and edit the menu. Administrator: all of these functions plus user management. Every authenticated user can change their own password.

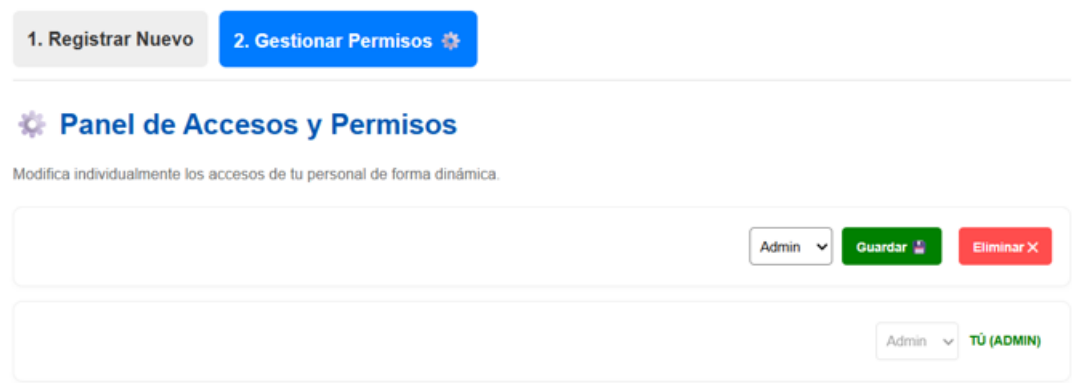


Figure 17. Access and Permissions panel.

3.8 Inventory Screen

Administrative screen with the “Raw Materials Manager” and “Purchase History” tabs. Requires the manage_inventory permission.

Raw materials manager

Groups base ingredients, toppings, and packaging. Allows users to create and edit the name, type, unit, current stock, and minimum stock; it displays low-stock or out-of-stock alerts. Users can hide/show items and open the restock or delete form.

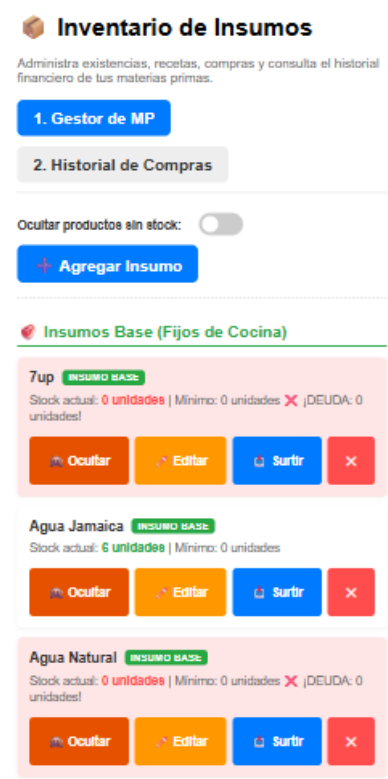


Figure 18. Raw Materials Manager.

Recipes and stock deductions

When editing a product, the administrator can associate inventory items and quantities with its recipe. When an order is submitted, a transaction deducts recipe quantities and selected ingredients from inventory when the corresponding inventory item exists. The “Hide out-of-stock products” toggle hides menu products whose recipe contains an out-of-stock item.

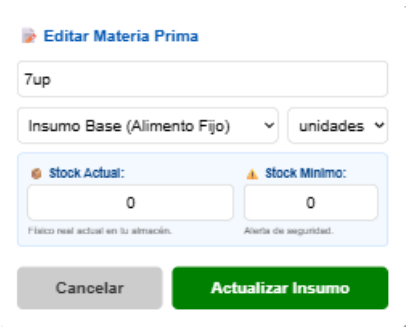
The screenshot shows a web form titled "Editar Materia Prima". It features a text input field containing "7up". Below it is a dropdown menu labeled "Insumo Base (Alimento Fijo)" and another dropdown labeled "unidades". A section with a light blue background contains two input fields: "Stock Actual:" with the value "0" and "Stock Mínimo:" with the value "0". Small text below these fields reads "Físico real actual en la almacén." and "Alerta de seguridad." respectively. At the bottom are two buttons: a grey "Cancelar" button and a green "Actualizar Insumo" button.

Figure 19. Edit Raw Material.

Purchase records

The restock form requests the date, quantity, cost, brand, and supplier. When saved, it adds the quantity to stock and records the transaction in purchases. The Purchase History tab lists the date, item, type, previous and current stock, quantity, unit, cost, brand, and supplier.

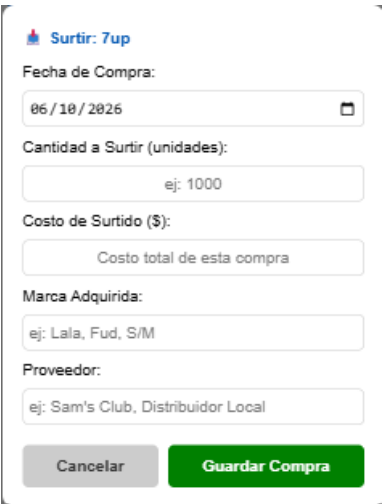
The screenshot shows a web form titled "Surtir: 7up". It includes a date picker for "Fecha de Compra:" set to "06/10/2026". Below is a text input for "Cantidad a Surtir (unidades):" with the value "ej: 1000". This is followed by a text input for "Costo de Surtido (\$):" with the placeholder "Costo total de esta compra". Then there is a text input for "Marca Adquirida:" with the value "ej: Lala, Fud, S/M". Finally, a text input for "Proveedor:" with the value "ej: Sam's Club, Distribuidor Local". At the bottom are two buttons: a grey "Cancelar" button and a green "Guardar Compra" button.

Figure 20. Restock Raw Materials.

Add Inventory Item

Registers a new raw material for use as an ingredient in recipes or as an inventory item for another type of product. The form requests the raw-material name, its item type (Base ingredient, selectable ingredient, or packaging/disposable), and its unit of measure (grams, milliliters, ounces, or units).

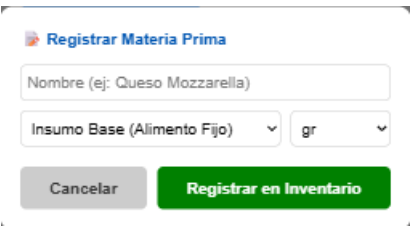
The screenshot shows a web form titled "Registrar Materia Prima". It has a text input field for "Nombre (ej: Queso Mozzarella)". Below it is a dropdown menu for "Insumo Base (Alimento Fijo)" and another dropdown for unit of measure, currently showing "gr". At the bottom are two buttons: a grey "Cancelar" button and a green "Registrar en Inventario" button.

Figure 21. Register Raw Material.

4. Cross-Functional Business Logic

4.1 Pricing Rules

Table 2. Pricing rules by product, order type, and number of ingredients.

Product / case	Pricing rule
Pizza Dog — Dine in	\$60 per unit, no promotion, regardless of quantity.
Pizza Dog — Takeout	Pairs cost \$100 (\$50 each); any leftover odd unit costs \$60.
Customizable product	Base price (set per product) + (number of ingredients – 1) × extra-ingredient price.
Extra-ingredient price	Configurable in Menu Settings; defaults to \$16 if no value is defined in Firestore.
Standard product	Fixed price defined in the product catalog.

4.2 Product Visibility and Scheduling

- Any product can be marked as hidden, independently of its dates, to temporarily remove it from the menu without deleting it.
- A product with a scheduled start date does not appear in the menu until that date and time.
- A product with an end (expiration) date automatically disappears when that date and time arrive, without manual intervention.

4.3 Identifier Generation

- Product ID:** generated automatically from the name (lowercase slug without accents, symbols, or linking words).
- Ingredient ID:** generated from the ingredient name in lowercase, with words joined by underscores.
- Order ID: format NO-YY-DDD-NNN, with a daily sequence incremented using a Firestore transaction (see Section 2.3).

4.4 Roles and Access

Each user has different interface permissions to support navigation according to their responsibilities on the site.

Table 3. User roles and assigned permissions.

Permission or function	Waiter	Kitchen	Manager	Admin
Take orders	Yes	No	Yes	Yes
Dispatch orders	Yes	Yes	Yes	Yes
Manage inventory	No	No	Yes	Yes
View sales	No	No	Yes	Yes
Edit menu	No	No	Yes	Yes
Manage users	No	No	No	Yes
Change own password	Yes	Yes	Yes	Yes

5. Data Model (Cloud Firestore)

Table 4. Collections, documents, and fields stored in Cloud Firestore.

Collection	Main fields
products	id, name, description, price, type, image, customizable, maxIngredients, temporary, start, end, label, specialType, hidden, order.
ingredients	id, name, hidden.
counters	day/prefix ID, last (daily order sequence).
settings/prices	extraIngredient, autoHideOutOfStock, transfer and WhatsApp details; may include menu ordering.

Collection	Main fields
orders	orderId, customer, orderType, items[] (quantity, name, details, total, orderType), total, paymentMethod, date, time, status.
users	name, email, role, permissions (take_orders, dispatch_orders, manage_inventory, view_sales, edit_menu).
inventory	name, type, unit, currentStock, minimumStock, recipe/visibility, and purchase data.
purchases	purchaseDate, rawMaterial, type, previousStock, restockedQuantity, currentStock, inventoryUnit, cost, brand, supplier, createdAt.

Technical note: the product “order” field and the screen for arranging products by numeric order are present in the code (manual menu reordering function), but they are not currently connected to a visible sidebar button. This functionality is documented as present in the logic while awaiting confirmation on whether it should be exposed in the interface.

6. Shared Elements

6.1 Footer

- Instagram link button.
- WhatsApp link button with a pre-filled greeting message.
- Copyright notice.

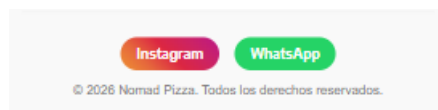


Figure 15. Footer (public and admin interfaces).

6.2 Responsive Design

- Mobile-first design: on narrow screens, product cards display the image above the content.
- At widths of 550px and above, cards switch to a horizontal layout (fixed-width image on the left, content on the right), including the Pizza Nomad/customizable pizza builder.
- Drink and pizza lists use grids that automatically adapt to the available width.